

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	29 June 2026
Team ID	
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Data Analysis Pipeline Flow

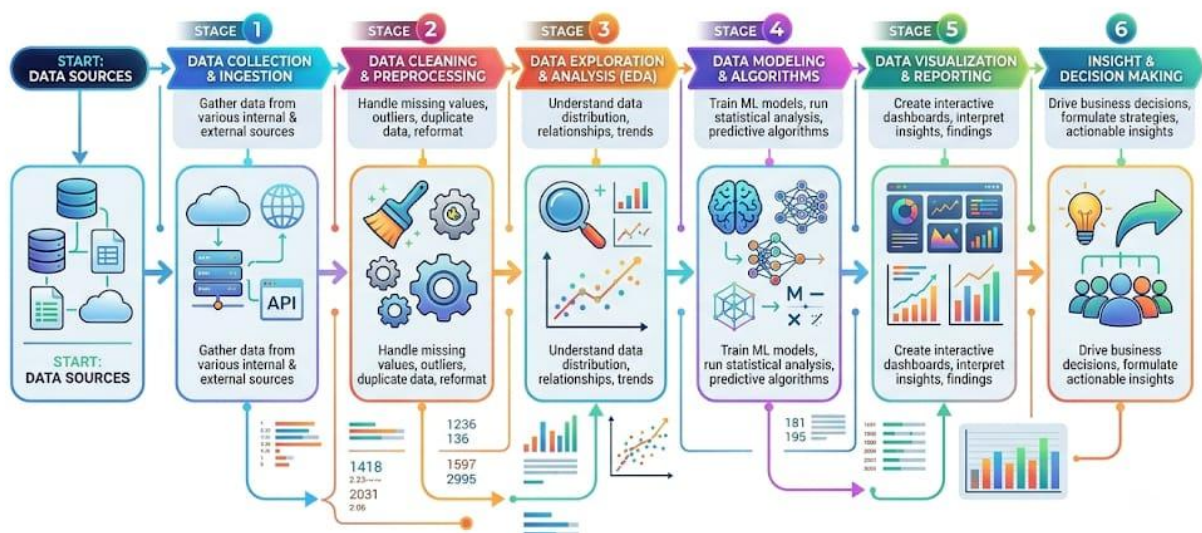


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Tableau Dashboard/ Storyboard	Tableau Desktop
2.	Data Cleaning	Preparing and transforming raw data	Tableau Prep / Python (Pandas)
3.	Data Source	Hosting UNESCO raw datasets	CSV / Excel / SQL Database
4.	Cloud Storage	Hosting the finished dashboard	Tableau Public / Cloud
5.	External API	Fetching real-time site data (if applicable)	UNESCO API / REST API

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Python (Pandas NumPy for data prep), Tableau Desktop
2.	Security Implementations	List all the security / access controls implemented	Tableau Public Row-Level Security, HTTPS/TLS encryption, Data masking for sensitive fields
3.	Scalable Architecture	Justify the scalability of architecture	Data Extracts for fast, scalable querying of large UNESCO datasets
4.	Availability	Justify the availability of application	Tableau Public Cloud infrastructure, CDN-backed delivery
5.	Performance	Design consideration for the performance	Data sources aggregation, using context filters, and optimized dashboard layout

References:

<https://whc.unesco.org/en/list/>

<https://help.tableau.com/>

https://help.tableau.com/current/prep/en-us/prep_welcome.htm